Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 | www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data and ML Engineer with 3 years of industry experience building production **Python** data pipelines, **PostgreSQL** data models and document-extraction services in regulated financial services, plus ML evaluation and data-preparation work from an MSc in Data Science (University of Bristol). Built a pipeline that extracts structured records from **300+** daily emails, a **Normalized SQL Schema** fed by API ingestion that merged in legacy data, and **RAG** services on **Azure OpenAI**. Co-built a preprocessing pipeline with an audit manifest for an MSc dissertation with AstraZeneca, and compared captioning models against **VLM** baselines. Acted as de facto team lead for up to 4 interns.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Data Engineering & Document Pipelines

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data previously held in Excel; developed **Optimized SQL** queries, **Functions** and **Procedures**.
- Built a **Python** extraction pipeline that parses over **300** daily emails from credit rating agencies into structured data (first with **AWS Textract**, then **Azure OpenAI** when an agency began sending emails as images), **fuzzy-matches** company names against the ones analysts track and sends a daily summary, **saving 60 analyst hours a month**; ran it as a scheduled batch job on **AWS SageMaker** through **EventBridge** and **Lambda**.
- Analyzed customer data with **Python Pandas** and **Market Basket Analysis** (**Apriori**) for the marketing team; migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**.

GenAI, NLP & Retrieval

- Designed and built **LLM applications** end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) for Compliance, Client Service and Investment, with **Azure OpenAI embeddings** and a separate **pgvector** database per department so each team searched only its own internal documents.
- Built **tool-calling** agents that call **Python** functions as tools to run **SQL**-backed fund comparisons and generate PDF reports on request; guided behaviour with **system prompts**, versioned every prompt change in **Git** and had the **Compliance** department review each release.
- Made cost-versus-quality model decisions: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and the stronger **Azure OpenAI** outputs for decision makers; released prototypes to real users and iterated on their feedback.

Security, Delivery & Stakeholders

- Worked under security and compliance constraints: **RBAC** with a **Python** authentication function against the company **LDAP** server, **AWS Security Groups** limiting access to company private IPs and an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

- Packaged the **FastAPI** microservices as **Docker** containers on **AWS** (**EC2**, **ECR**, **Lambda**) and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Worked directly with Client Service, Compliance and Investment from requirements to release; built the first systems as the only engineer alongside my manager, later acted as de facto team lead for up to 4 interns; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; personal focus on the data preprocessing, scoring algorithm and RAG assistant. Preprocessing pipeline that joins 17 raw and augmented inputs (DepMap, GEO, HPA) into 14 clean tables, with identity resolution across sources, integrity checks and a build manifest audit file; scoring ranks cancer cell lines by target inclusion and exclusion genes across six evidence layers and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Study comparing CNN/ViT/ResNet encoders with symbolic, SBERT embedding-retrieval and LSTM sequence decoders (with and without Bahdanau attention) for generating image descriptions on three synthetic datasets with train/validation/test splits, over five seeds, scored with SBERT cosine similarity, BLEU and ROUGE-L and compared with BLIP/SigLIP and commercial VLMs, noting where the evaluation protocols differed.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **Swindon Borough Council Fuel-Poverty Hack:** A household and neighborhood-based fuel poverty data analysis tool. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

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ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Azure OpenAI, LlamaIndex, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn, fuzzywuzzy), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite

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